

MODULE 02 · DOMAIN 1 - CLOUD CONCEPTS

# The Well-Architected Framework

Six pillars, and the one word in a question that picks between them.

DOMAIN 1 of 4   EXAM WEIGHT 24%   MODULE 2 of 26

The pillars are not a checklist. They are **six competing priorities** you trade against each other.

# The six pillars

| PILLAR                 | THE QUESTION IT ANSWERS                                                  |
|------------------------|--------------------------------------------------------------------------|
| Operational Excellence | Can we run, monitor and improve this system day to day?                  |
| Security               | Can we protect data, systems and assets, and prove who did what?         |
| Reliability            | Does it do its job correctly and consistently, and recover from failure? |
| Performance Efficiency | Are we using the right resources efficiently as demand changes?          |
| Cost Optimization      | Are we getting business value at the lowest price?                       |
| Sustainability         | Are we minimising the environmental impact of running this?              |

## ON THE EXAM

Six, not five. Sustainability was added in 2021 and is examinable. Note that AWS spells the pillar Cost Optimization.

# Reliability or Performance Efficiency?

## RELIABILITY

### Does it keep working?

- Recovers automatically from failure
- Performs its function correctly and consistently
- Multi-AZ, backups, health checks, retries
- Failure is the subject

## PERFORMANCE EFFICIENCY

### Does it work well enough, efficiently?

- Right resource type for the workload
- Adapts as demand and technology change
- Caching, serverless, appropriate instance types
- Efficiency is the subject

## THE TIEBREAKER

If the scenario mentions recovering from failure, it is Reliability. If it mentions using resources efficiently, it is Performance Efficiency.

# Operational Excellence or Reliability?

## OPERATIONAL EXCELLENCE

### How the team runs it

- Deployment processes and runbooks
- Observability and small reversible changes
- Learning from operational failures
- The people and process side

## RELIABILITY

### How the system behaves

- Automatic recovery from failure
- Scaling horizontally to reduce blast radius
- Testing recovery procedures
- The architecture side

## THE TIEBREAKER

Operational Excellence is about the practice of running the system. Reliability is about the system surviving without you.

# Design principles worth knowing by name

- ◆ **Stop guessing capacity.** Provision to measured demand.
- ◆ **Test systems at production scale.** Cheap; you delete the environment after.
- ◆ **Automate to make experimentation easier.** IaC makes changes reversible.
- ◆ **Allow for evolutionary architectures.** Design so it can change.
- ◆ **Drive architectures using data.** Measure, then decide.
- ◆ **Improve through game days.** Rehearse failure deliberately.

## ON THE EXAM

Game days and testing at production scale are the two that appear as distractor-resistant correct answers.

# The Well-Architected Tool

- ◆ **AWS Well-Architected Tool** is a free service in the console that reviews a workload against the pillars.
- ◆ You answer a structured questionnaire per workload and it produces an improvement plan.
- ◆ It is not continuous. It is a review you run, not a monitor that watches.
- ◆ Contrast with AWS Trusted Advisor, which inspects your live account continuously and automatically.
- ◆ Lenses adapt the review to a specific workload type, such as serverless or machine learning.

## ON THE EXAM

Continuous and automatic means Trusted Advisor. A structured review you sit down and complete means the Well-Architected Tool.

# Pillar reverse lookup

| IF THE SCENARIO SAYS                                   | THE PILLAR IS          |
|--------------------------------------------------------|------------------------|
| Recover quickly from failure, consistently correct     | Reliability            |
| Protect data, control access, trace actions            | Security               |
| Run and monitor day to day, improve processes          | Operational Excellence |
| Use computing resources efficiently as demand changes  | Performance Efficiency |
| Avoid unnecessary spend, deliver value at lowest price | Cost Optimization      |
| Reduce energy use and environmental impact             | Sustainability         |

## ON THE EXAM

The exam almost always uses these exact phrasings. Learn the phrasings, not just the pillar names.



A payments company is redesigning its transaction service. The requirement is that the service must continue processing correctly if an entire Availability Zone becomes unavailable, and must return to full capacity without an engineer intervening.

The team is deploying across three Availability Zones with health checks and automatic replacement of failed instances.

**Which pillar of the Well-Architected Framework does this design represent?**

- |                                                   |                                                   |
|---------------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> A Security               | <input type="checkbox"/> B Performance Efficiency |
| <input type="checkbox"/> C Operational Excellence | <input type="checkbox"/> D Reliability            |

◆ PAUSE AND DECIDE

## Which pillar of the Well-Architected Framework does this design represent?

**A** Security

*Payments implies security matters, but no access control, encryption or audit requirement is stated.*

**C** Operational Excellence

*Close, but the recovery is automatic. Operational Excellence would be about the team's processes and runbooks.*

**B** Performance Efficiency

*Nothing about resource choice or efficiency. The design is about surviving loss, not running leanly.*

**D** Reliability

*Continues correctly through failure, recovers without intervention. That is the pillar definition.*

### THE SIGNAL

Two phrases: continue processing correctly, and return to full capacity without intervention.

A research group runs nightly simulations on a fixed fleet of large general-purpose EC2 instances. The jobs are heavily compute-bound, and profiling shows memory sits at eleven percent while CPU pins at one hundred percent for hours.

The team wants the jobs to finish sooner without increasing spend, and is evaluating compute-optimised instance types and a caching layer for the shared input data.

**Which pillar is the team primarily working in?**

- |                                              |                                                   |
|----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> A Reliability       | <input type="checkbox"/> B Performance Efficiency |
| <input type="checkbox"/> C Cost Optimization | <input type="checkbox"/> D Sustainability         |

◆ PAUSE AND DECIDE

## Which pillar is the team primarily working in?

- A

 Reliability  
*Nothing has failed and nothing needs to recover. The jobs complete, just slowly.*
- B

 Performance Efficiency  
*Selecting the right resource type for the workload and adding caching is the pillar's definition.*
- C

 Cost Optimization  
*The strongest distractor. But spend is a constraint here, not the goal. The goal is finishing sooner.*
- D

 Sustainability  
*Efficient use does reduce energy, but no environmental goal is stated. Do not infer one.*

### THE SIGNAL

Without increasing spend is a constraint. Finish sooner is the goal. The goal picks the pillar.

An operations manager wants to reduce the number of production incidents caused by deployments. The team currently ships one large release a month, has no rollback procedure, and diagnoses problems by reading application logs on individual servers.

The proposed changes are: deploy small changes frequently, script the rollback, centralise logs and metrics, and hold a review after every incident.

**Which pillar do these changes belong to?**

☐ A Operational Excellence

☐ B Reliability

☐ C Security

☐ D Performance Efficiency

◆ PAUSE AND DECIDE

## Which pillar do these changes belong to?

**A** Operational Excellence

Small frequent changes, scripted rollback, observability and learning from incidents are its four design principles.

**C** Security

Centralised logs support security, but the stated purpose is diagnosing deployments, not detecting threats.

**B** Reliability

*Genuinely tempting, since fewer incidents means more uptime. But nothing changes the architecture, only the practice.*

**D** Performance Efficiency

*No resource selection or efficiency question anywhere in the scenario.*

THE SIGNAL

Every proposed change is a process change. Nothing about the system's architecture moves.

A company wants a structured review of its main workload against AWS best practices. It wants to answer a set of questions about the workload, receive a prioritised list of improvements, and repeat the review each quarter to track progress.

It also wants automatic, ongoing checks of the live account for unused resources and open security groups, without anyone completing a questionnaire.

**Which two services meet these requirements? (Choose two.)**

- ☐ A AWS Well-Architected Tool
- ☐ B AWS Trusted Advisor
- ☐ C AWS Config
- ☐ D AWS Compute Optimizer
- ☐ E Amazon Inspector

◆ PAUSE AND DECIDE

## Which two services meet these requirements? (Choose two.)

- A** AWS Well-Architected Tool  
The questionnaire-based, repeatable workload review producing an improvement plan.
- B** AWS Trusted Advisor  
Continuous automatic checks including idle resources and open security groups.
- C** AWS Config  
*Records configuration and evaluates compliance rules. It does not recommend best practices unprompted.*
- D** AWS Compute Optimizer  
*Rightsizing recommendations only. Narrower than the account-wide checks described.*
- E** Amazon Inspector  
*Scans workloads for software vulnerabilities and unintended network exposure, not cost or best practice.*

### THE SIGNAL

Two requirements, two services. Questionnaire means Well-Architected Tool. Continuous and automatic means Trusted Advisor.



# What to carry out of this module

- ◆ **Six pillars.** Sustainability is the sixth and it is examinable.
- ◆ **Reliability worries about failure. Performance Efficiency worries about efficiency.**
- ◆ **Operational Excellence is process. Reliability is architecture.** Ask where the fix lives.
- ◆ The goal picks the pillar, not the constraint. Without increasing spend is a constraint.
- ◆ Well-Architected Tool is a review you run. Trusted Advisor watches continuously.

## ON THE EXAM

AWS spells the pillar Cost Optimization with a z. It is a proper noun, so use the AWS spelling in exam contexts.